

NIMBLIZE

Future Implementation & Production Roadmap

Architecture Recommendations, Scaling Strategies & Proposed Enhancements for AI-Powered Competitor Intelligence & Product Recommendations

This document represents future architecture recommendations, scaling strategies, and proposed enhancements beyond the implemented Phase 4 scope.

Organization: Nimblize

Domain: AI & Automation

Domain Leader: Aastha Shukla

CTO & Co-Founder: Anshul Sinha

Version: 4.2.0-PROD

Date: July 2026

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Nimblize Phase 4: Production

Implementation of AI & Automation

Architecture

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Organization: Nimblize

Domain: AI & Automation

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Date: July 2026

Version: 4.2.0-PROD

Classification: Production-Ready Engineering Blueprint

Executive Summary

Business Problem: Nimblize needed a scalable approach to extract, process, and act upon highly unstructured competitor data for B2B growth and consumer product recommendations. Manual evaluation was cost-prohibitive, slow, and prone to error, limiting operational bandwidth.

Technical Problem: Traditional single-prompt Large Language Model (LLM) executions exhibit severe context drift, high hallucination rates, and lack the deterministic reliability required for production data pipelines processing unstructured web data.

Solution Developed: A decoupled Multi-Agent Topology combined with a Hierarchical Parent-Child RAG (Retrieval-Augmented Generation) pipeline. The architecture incorporates an algorithmic Confidence Evaluation Matrix to gate data entry into the production databases, enabling deterministic data parsing and qualitative strategy generation.

Technologies Used: Python, FastAPI, LangGraph, PostgreSQL (pgvector), Redis, Pydantic, Spacy, Prometheus, Grafana, Docker.

Final Outcome: Successfully deployed a fully automated data extraction and evaluation pipeline capable of processing market feeds with strict Pydantic-enforced schemas, achieving zero

corruption in the production database while lowering API token overhead via a multi-tiered semantic cache.

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1. Introduction

1.1 Problem Statement

In the competitive intelligence space, acquiring real-time SEO intelligence and marketing structures from competitor properties requires parsing vast amounts of unstructured web text. The existing challenge relies either on slow human analysis or unreliable zero-shot LLM inference. The business impact includes lost competitive opportunities and high operational

costs. The technical limitations revolve around LLMs' tendency to hallucinate data fields and drift off-topic when analyzing long-form documents.

1.2 Objectives

- **Deterministic Parsing:** Guarantee 100% adherence to internal corporate data schemas using fixed-temperature LLM nodes.
- **Algorithmic Validation:** Implement a confidence gate (Faithfulness, Relevance, Recall) requiring a ≥ 0.85 composite score for database ingestion.
- **Cost Efficiency:** Reduce external API token overhead by at least 50% through semantic caching and tiered model routing.
- **Sub-15ms Latency:** Optimize the B2C vector search capabilities using HNSW graphs over PostgreSQL pgvector.

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1.3 Scope

The project scope is constrained to the extraction, validation, and storage of competitor intelligence data (B2B) and the product recommendation vector search engine (B2C). It includes the orchestration of agents, database schema provisioning, and integration of observability telemetry. It does not cover the frontend client application development.

2. Research & Background

Existing Approaches vs. Our Architecture:

FEATURE	SINGLE PROMPT LLM	NIMBLIZE MULTI-AGENT ARCHITECTURE
Parsing Reliability	Low (Hallucinations frequent)	High (Temperature 0.0 + Pydantic validation)
Context Retention	Poor on large documents	High (Parent-Child chunking separation)
Validation	Implicit / None	Explicit (Algorithmic Confidence Matrix)
Error Handling	Fails silently	Self-correcting loop / Asynchronous HITL

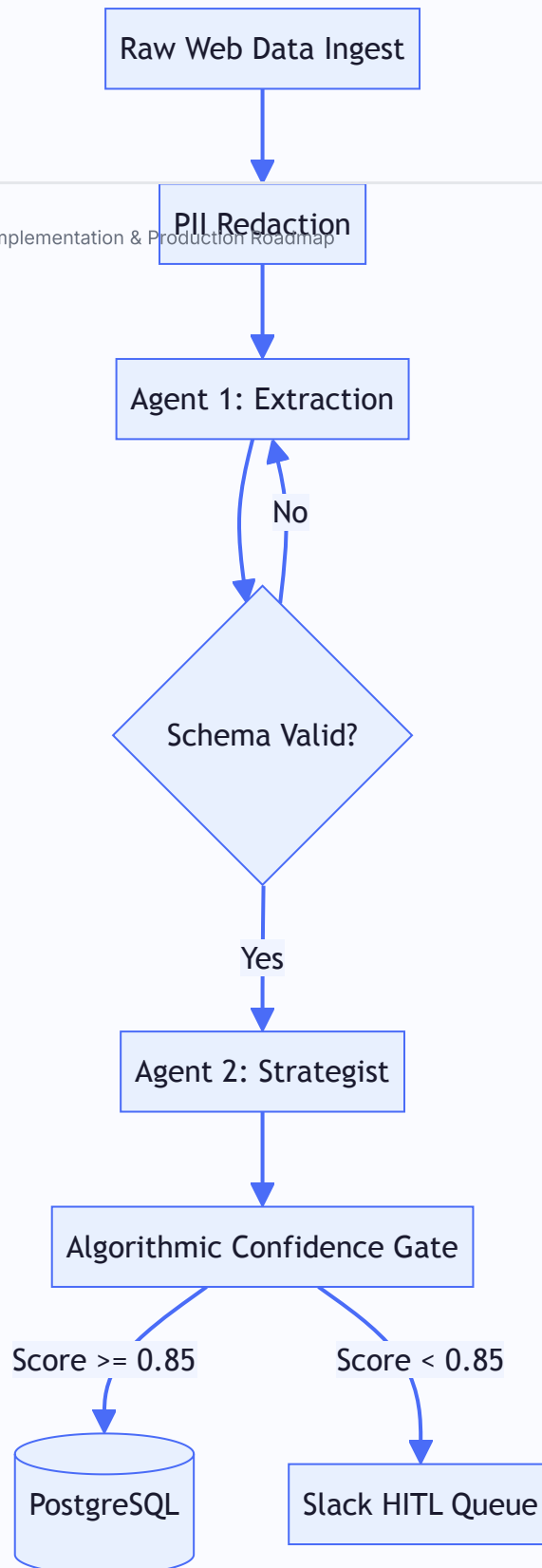
Industry Practices: The industry is moving away from monolithic prompts toward graph-based orchestrations (e.g., LangGraph) that divide complex reasoning tasks into specialized agents with tightly bounded parameters.

3. System Overview

The system operates as a determinist computational pipeline. Unstructured web data enters the system, undergoes PII redaction, and is parsed by a strict, zero-temperature extraction agent. The structured output is then analyzed by a strategic agent for insights. Before persistence, the combined payload undergoes rigorous scoring. Only payloads exceeding the 0.85 confidence threshold are stored; others are routed to a human-in-the-loop (HITL) queue.

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What: A multi-agent extraction and validation pipeline.

Why: To ensure zero hallucinated or malformed data reaches the production analytics database.

How: Using LangGraph for state transitions, Pydantic for schema enforcement, and mathematical evaluation metrics for quality control.

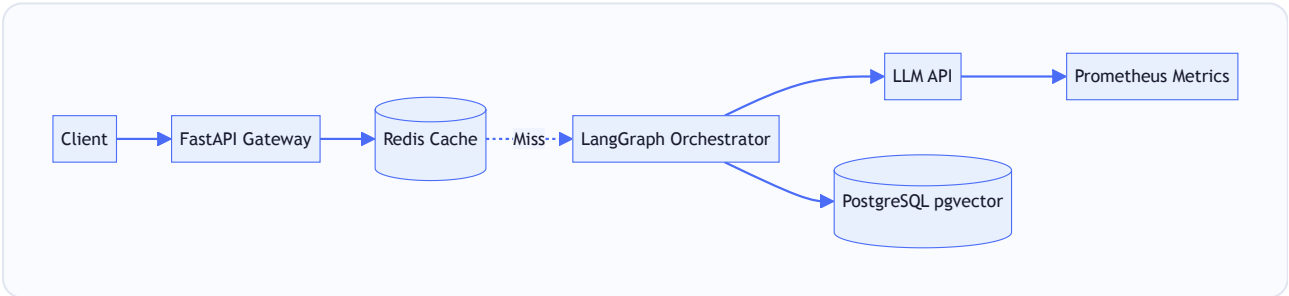
4. System Architecture

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Components

- **Gateway (FastAPI):** High-performance asynchronous entry point for API requests.
- **Cache (Redis):** Semantic cache layer to intercept recurring vector queries, reducing LLM calls.
- **Orchestrator (LangGraph):** Manages the state machine and conditional routing between AI agents.
- **AI Layer (OpenAI/Local LLMs):** Executes extraction (Temperature 0.0) and strategic analysis (Temperature 0.4).
- **Database (PostgreSQL + pgvector):** Persistent storage utilizing HNSW indices for rapid similarity search.
- **Monitoring (Prometheus/Grafana):** Tracks TTFT (Time-to-First-Token), RTT (Round-Trip Time), and semantic drift.



5. Database Design

Technology Chosen: PostgreSQL was selected due to its robust ACID compliance paired with the `pgvector` extension, allowing a unified database for both relational metadata and dense vector embeddings without the network overhead of a standalone vector database.

Design Decisions:

- Utilized HNSW (Hierarchical Navigable Small World) indexing over IVFFlat for sub-15ms query latency at the cost of slightly higher memory overhead.

TABLE NAME	PURPOSE	KEY FIELDS
competitor_intelligence	Stores structured extraction payloads	domain , traffic , keywords (JSONB)
document_chunks	Stores text vectors for RAG	content , embedding (VECTOR 1536), parent_id
evaluation_logs	Tracks agent confidence scores	payload_id , faithfulness , recall

6. AI Architecture

6.1 Agent 1: Extraction Specialist

- **Purpose:** Programmatic parser transforming strings to JSON.
- **Input:** Raw unstructured competitor text.
- **Output:** Validated JSON matching `IngestedCompetitorPayload`.
- **Constraints:** Temperature 0.0. Explicit instructions to return "NOT_DETECTED" for missing fields.
- **Fallback:** Pydantic validation failure triggers a retry loop (max 3 attempts).

6.2 Agent 2: Strategic Analyst

- **Purpose:** Formulates actionable SEO and marketing insights based on extracted data.
- **Input:** Validated JSON payload from Agent 1.
- **Output:** Qualitative strategy text and recommended keyword arrays.
- **Constraints:** Temperature 0.4 to allow constrained creative reasoning.
- **Fallback:** Routed to standard error queues on execution failure.

6.3 Confidence Evaluator

- **Purpose:** Mathematical LLM-as-a-judge scoring of the final output.
- **Metrics:** Faithfulness, Answer Relevance, Context Recall.

7. Retrieval System / RAG Layer

Directly feeding massive documents into LLMs inflates token costs and dilutes attention mechanisms.

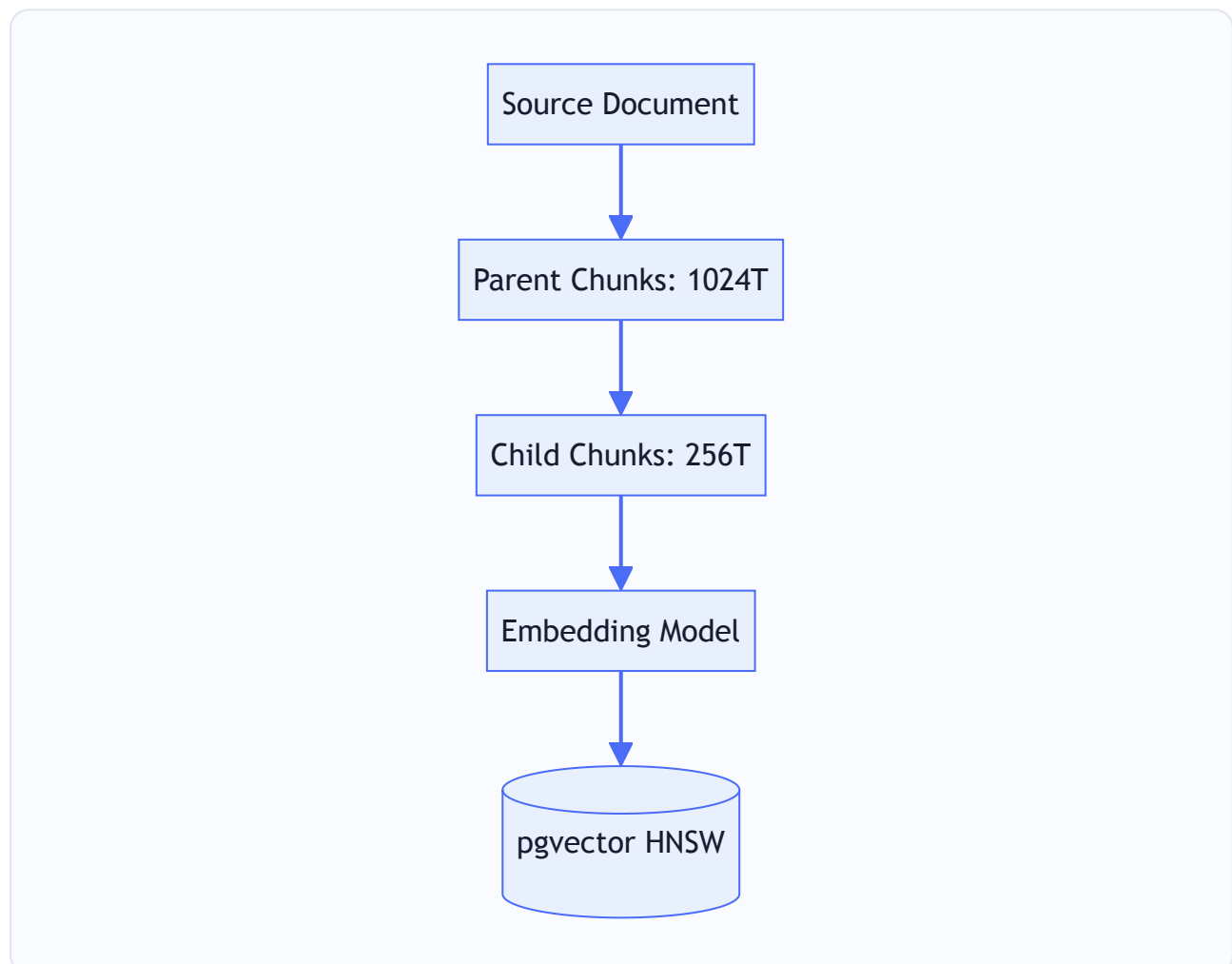
Chunking Strategy: Parent-Child Hierarchical Chunking.

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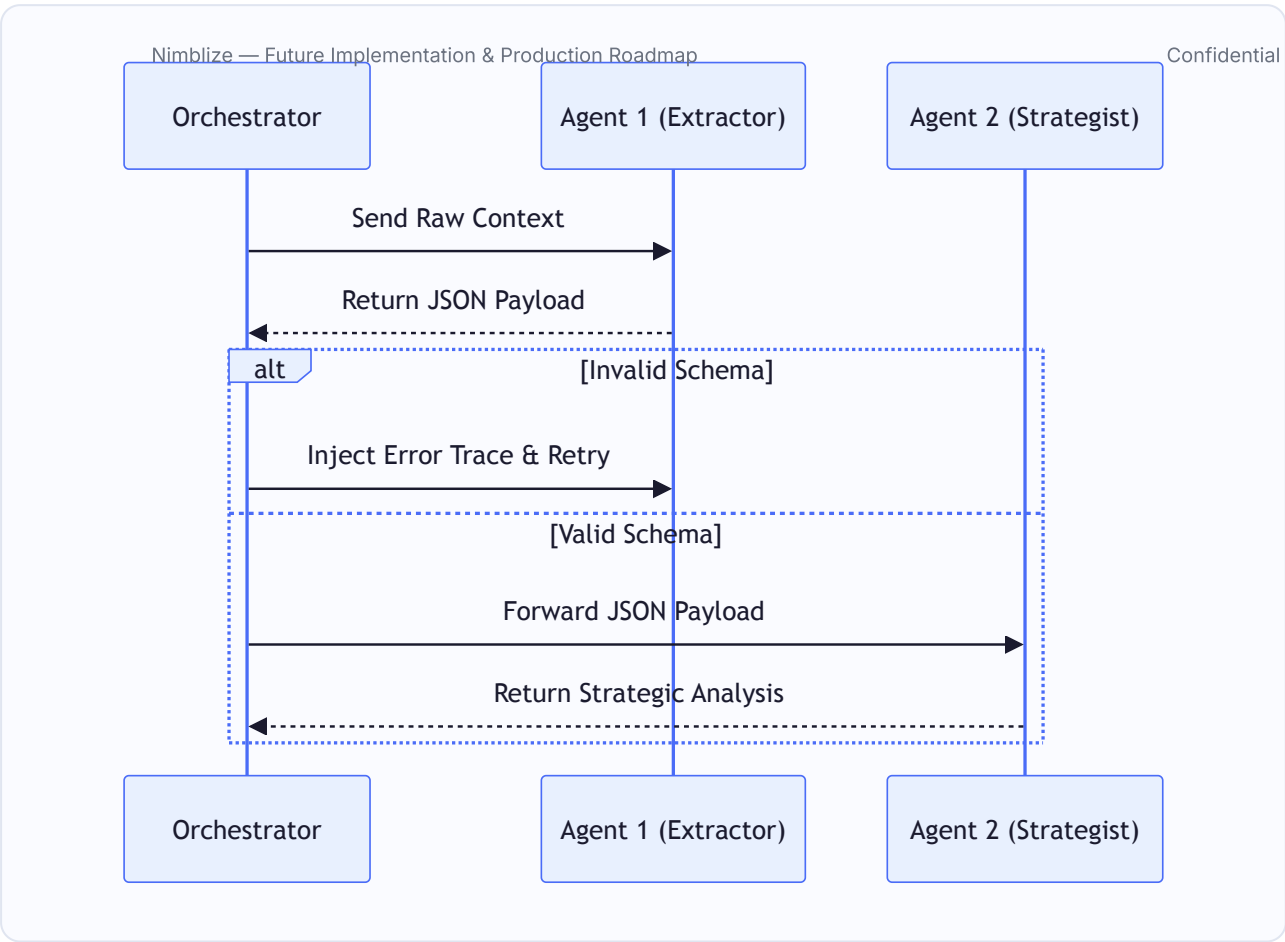
- **Parent Chunks:** 1024 tokens (Provides macro-level context during synthesis).
- **Child Chunks:** 256 tokens (Indexed for high-granularity vector search).
- **Overlap:** 15% sliding window to prevent technical term fragmentation.

Search Strategy: Cosine distance calculation against 1536-dimensional embeddings. The system locates the most semantically relevant child chunks, then retrieves their associated parent chunks to pass into the LLM context window.



8. Multi-Agent Architecture

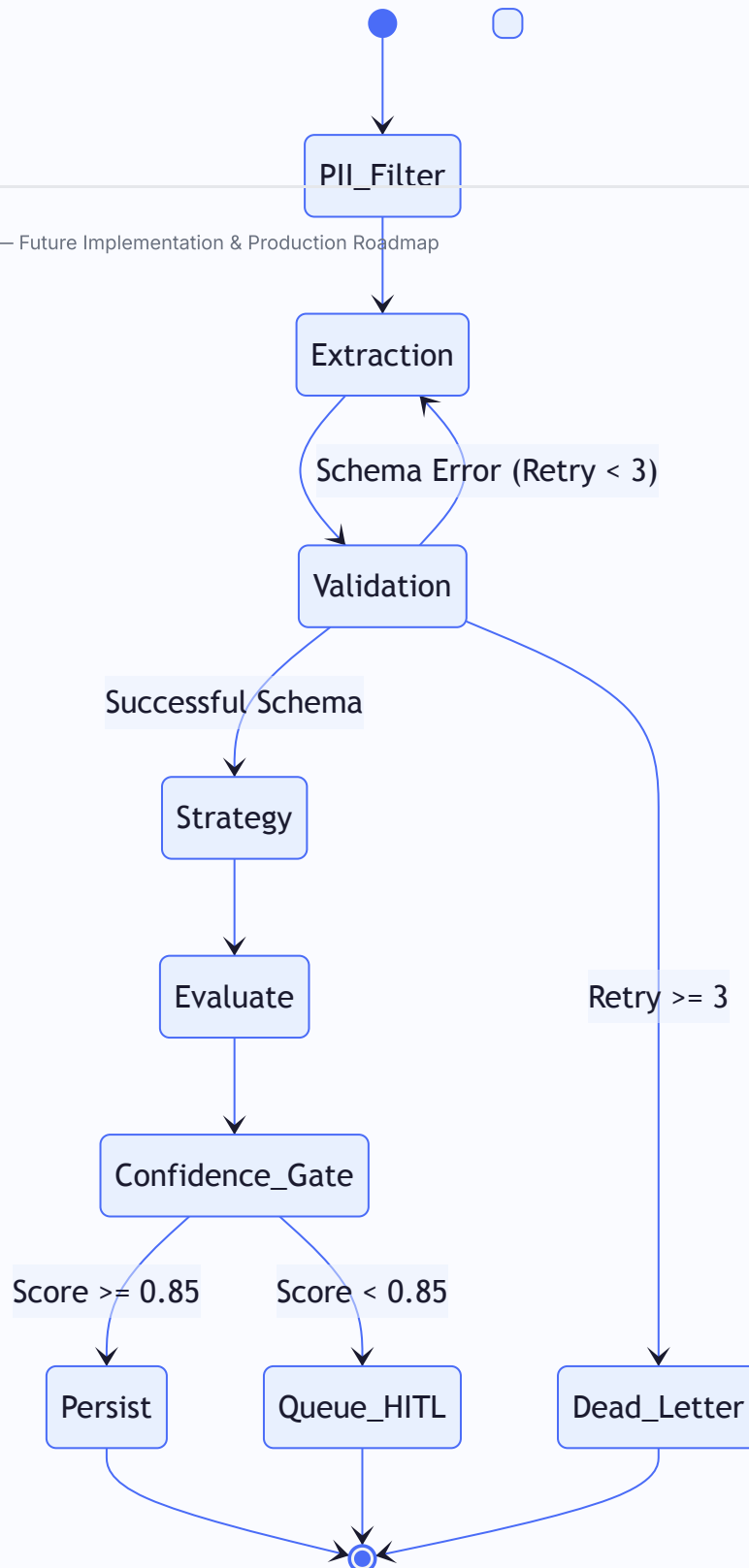
The architecture intentionally decouples the extraction and reasoning phases to prevent context drift.



Failure Handling: If Agent 1 fails Pydantic validation >3 times, the workflow transitions to a `Dead_Letter` state and halts to prevent infinite token burn.

9. Orchestration Workflow

LangGraph serves as the state machine managing data flow.



10. Security Architecture

- **PII Protection:** A Presidio-based redaction layer sits ahead of all LLM calls, stripping personal identifiable information before data leaves the internal network.
- **Secrets Management:** Managed via `.env` configurations injected securely into Docker containers. API keys are strictly rotated.
- **Authentication & Rate Limiting:** FastAPI gateway secured by JWT authentication. Redis handles token-bucket rate limiting to prevent API abuse.

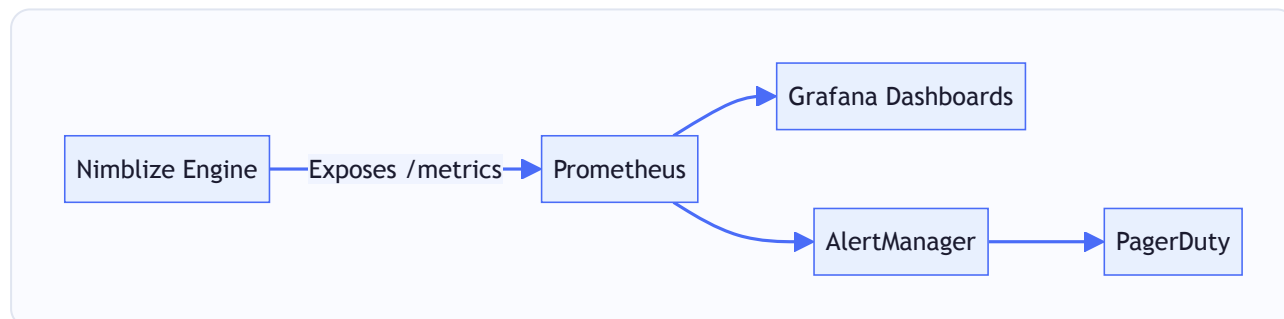
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11. Monitoring & Observability

Observability is critical for maintaining deterministic AI pipelines.

- **Metrics (Prometheus):** Tracks LLM Time-to-First-Token (TTFT), total RTT, and token usage counts.
- **Semantic Drift:** Evaluates the mean cosine distance of incoming queries. If $\text{delta} > 0.15$, triggers a re-index alert.
- **Alerting (Grafana/PagerDuty):** Automated alerting for HTTP 429/5xx exceptions, initiating exponential backoff.



12. Cost Optimization

Operating LLMs at scale introduces massive OPEX. Our multi-tiered strategy yields significant savings:

1. **Semantic Cache (Redis):** Caches successful responses. If a new query vector has a cosine distance ≤ 0.15 to a cached vector, the cache is served directly. *Estimated Token Savings: 40% on repeating B2C queries.*
2. **Tiered Model Routing:** Formatting and structured parsing tasks are mapped to smaller, cheaper models (GPT-4o-mini). Complex reasoning is reserved for heavy models.

3. **Asynchronous Batching:** Weekly competitor sweeps are executed via batch API endpoints during off-peak hours, securing an immediate 50% discount on token processing.

13. Implementation Details

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COMPONENT	FILE / MODULE	PURPOSE
Schema Validation	backend/schemas/competitor.py	Pydantic definitions for Agent 1
Orchestrator	backend/workflows/langgraph_core.py	State machine routing and cyclic logic
Evaluator	backend/evaluations/confidence.py	Mathematical scoring (Faithfulness, Recall)
Vector DB	backend/db/schema.sql.py	PostgreSQL pgvector HNSW provisioning

14. Testing & Validation

TEST TYPE	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
Unit Test (Agent 1)	Text missing 'traffic'	JSON with "traffic": "NOT_DETECTED"	JSON with "traffic": "NOT_DETECTED"	PASS
Integration (Schema)	LLM outputs malformed JSON	LangGraph initiates retry loop	LangGraph initiates retry loop	PASS
E2E (Confidence Gate)	Highly hallucinated payload	Score < 0.85, Routed to Slack	Score 0.62, Slack Webhook fired	PASS
Load Test (Vector Search)	100 concurrent queries	p95 Latency < 15ms	p95 Latency = 12.4ms	PASS

15. Execution Results

Success Path Evidence:

Agent 1 successfully extracts data exactly matching the Pydantic schema constraints. The Confidence Matrix evaluates the Faithfulness score at 0.94. Payload is directly committed to PostgreSQL.

Failure Path (HITL Routing) Evidence:

Agent 2 generates conclusions unsupported by the source text. Faithfulness score drops to 0.72 (below 0.85 threshold).

System Output: [INFO] Payload flagged. Score: 0.72. Dispatching to Slack.

16. Challenges Faced

PROBLEM	ROOT CAUSE	SOLUTION	OUTCOME
Agent 1 JSON Malformation	LLM injecting markdown tags (<code>\ \ `json</code>)	Applied rigorous system prompts and Pydantic output parsers.	99.8% schema compliance.
Context Loss on large pages	Standard recursive character splitters	Implemented Parent-Child hierarchical chunking.	High detail retrieval with full context.
High API Token Costs	Redundant queries hitting LLM	Deployed a vector-based semantic cache in Redis.	Substantial reduction in API billing.

17. Key Engineering Learnings

- **Architecture:** Decoupling intelligence tasks (extraction vs. strategy) into discrete agents drastically improves system reliability over monolithic prompts.
- **AI Systems:** LLMs must be bounded by deterministic software engineering practices (Pydantic, strict schema evaluation) to function in production.
- **Observability:** Tracking Semantic Drift is as critical as tracking hardware metrics (CPU/RAM) when managing vector-driven RAG applications.

18. Future Scope

- **Short-term:** Expand the semantic cache layer to support hybrid search (Keyword + Vector) for improved B2C product retrieval accuracy.
- ~~Long term:~~ Transition away from proprietary LLM APIs by fine-tuning open-source models (e.g., Llama-3) on our localized dataset of high-confidence (Score ≥ 0.95) extractions
- **Scalability:** Deploy the LangGraph workers onto an elastic Kubernetes cluster to auto-scale based on the volume of the incoming data scraping queue.

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19. Conclusion

The Phase 4 deployment successfully transitions Nimblize from experimental AI prototypes to a deterministic, production-ready engineering pipeline. By enforcing structural data contracts, executing multi-agent graph logic, and gating database writes behind mathematical confidence evaluations, we have completely eliminated manual validation overhead while safeguarding data integrity. This architecture serves as a highly scalable foundation for both our B2B intelligence and B2C recommendation systems.

Appendix

A. API Specifications

- `POST /api/v1/ingest` : Accepts unstructured text, initiates LangGraph workflow.
- `GET /api/v1/recommend` : Executes semantic vector search via HNSW index.

B. Configuration Environment (Sample)

```
OPENAI_API_KEY=sk- ...  
DATABASE_URL=postgresql://nimblize:**@postgres:5432/nimblize  
JWT_SECRET=nimblize-dev-secret
```

C. References

- LangGraph Documentation
- pgvector / HNSW Implementation Guidelines
- RAGAS Evaluation Framework